



Cambridge International AS & A Level

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MATHEMATICS

9709/35

Paper 3 Pure Mathematics 3

May/June 2025

1 hour 50 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

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[illegible]

2 Solve the equation $3 \cot \theta - 4 \operatorname{cosec}^2 \theta + 5 = 0$ for $-\pi \leq \theta \leq \pi$. [5]

[illegible]



3 The complex numbers s and t are given by

$$s = 5(\cos 0.25 + i \sin 0.25) \quad \text{and} \quad t = 6e^{3i}.$$

(a) Express $\frac{s}{t}$ in the form $re^{i\theta}$, where $-\pi < \theta \leq \pi$ and $r > 0$. [2]

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(b) In an Argand diagram with origin O , the points A and B represent the complex numbers s and $\frac{s}{t}$ respectively.

By considering the line segments OA and OB , or otherwise, state the two geometric effects of dividing a complex number by $6e^{3i}$. [2]

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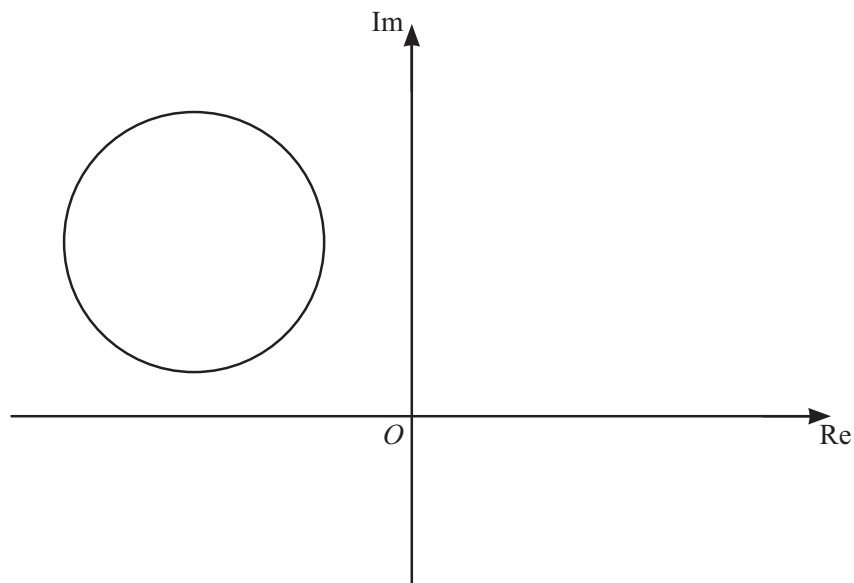
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4 Find the exact coordinates of the stationary point of the curve with equation $y = 3x^3 \ln x^4$, for $x > 0$. [5]

[illegible]



The diagram shows the locus of points representing the complex numbers, z , satisfying $|z + 5 - 4i| = 3$.

- (a) For the points on this locus, determine the maximum and minimum possible values of $|z|$. [3]

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- (b) For the points on this locus, determine the minimum possible value of $\arg z$. [3]

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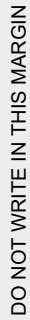
6 The parametric equations of a curve are

$$x = \frac{2}{\cos 3t} \quad \text{and} \quad y = \tan 3t,$$

for $0 \leq t \leq 2\pi$.

(a) Show that $\frac{dy}{dx}$ can be written as $A \operatorname{cosec} 3t$, where A is a constant to be found. [5]

[illegible]



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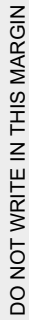
[illegible]



7 The equation of a curve is $y = \tan^{-1}(4x)$.

(a) Find the exact values of x when the gradient of the curve is $\frac{1}{4}$. [3]

[illegible]



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- 8 (a) By sketching a suitable pair of graphs, show that the equation $\sec 2x = -2x - \frac{1}{2}$ has exactly one root in the interval $0 \leq x \leq \frac{1}{2}\pi$. [2]

- (b) Show by calculation that this root lies between 0.8 and 1.2. [2]

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- (c) Show that, if a sequence of real values given by the iterative formula

$$x_{n+1} = \frac{1}{2} \cos^{-1} \left(\frac{-2}{4x_n + 1} \right)$$

converges, then it converges to the root of the equation in part (a).

[2]

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- (d) Use this iterative formula to calculate this root correct to 3 decimal places. Give the result of each iteration to 5 decimal places.

[3]

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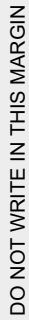


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10 With respect to the origin O , the points A , B and C have position vectors given by

$$\overrightarrow{OA} = 2\mathbf{i} - \mathbf{j} - 6\mathbf{k}, \quad \overrightarrow{OB} = b\mathbf{i} - 2\mathbf{j} + 3\mathbf{k} \quad \text{and} \quad \overrightarrow{OC} = -4\mathbf{i} + 5\mathbf{j} - 2\mathbf{k}.$$

(a) It is given that $|\overrightarrow{AB}| = |\overrightarrow{BC}|$.

Find the value of b .

[3]

[illegible]



- (b) A , B , C and D are the vertices of a rhombus.

Find the position vector of D .

[2]

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- (c) Calculate angle ABC .

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- 11** The variables x and y satisfy the differential equation

$$(x^2 + 3) \frac{dy}{dx} = e^{3y} (x - 2).$$

It is given that $y = 0$ when $x = 0$.

Solve the differential equation, and find the value of y when $x = 2$.

[8]

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